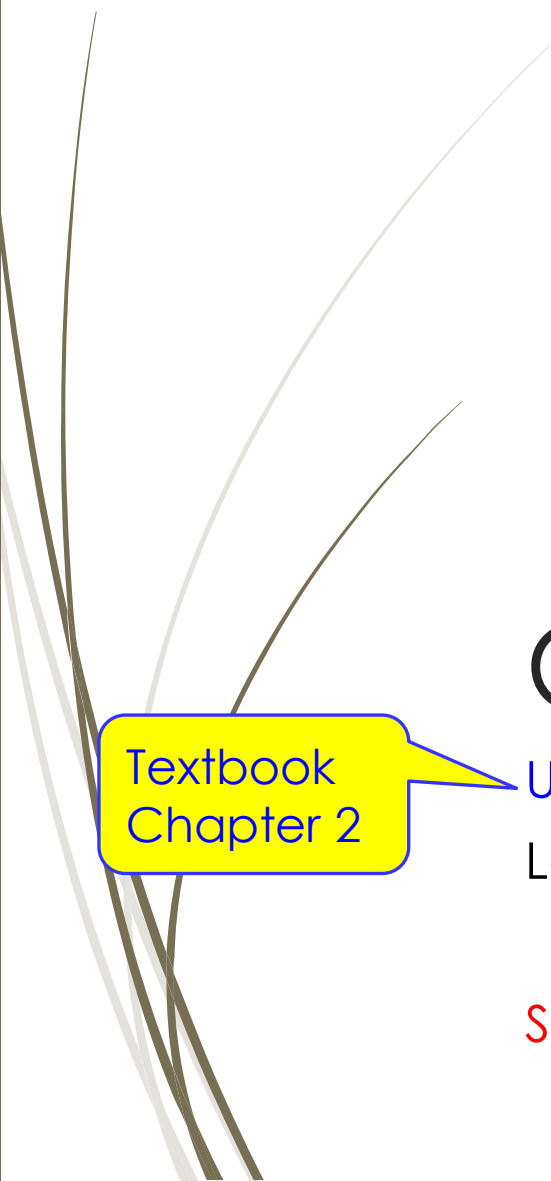




Textbook
Chapter 2

CMPT 295

Unit - Data Representation

Lecture 2 – Storing data in memory

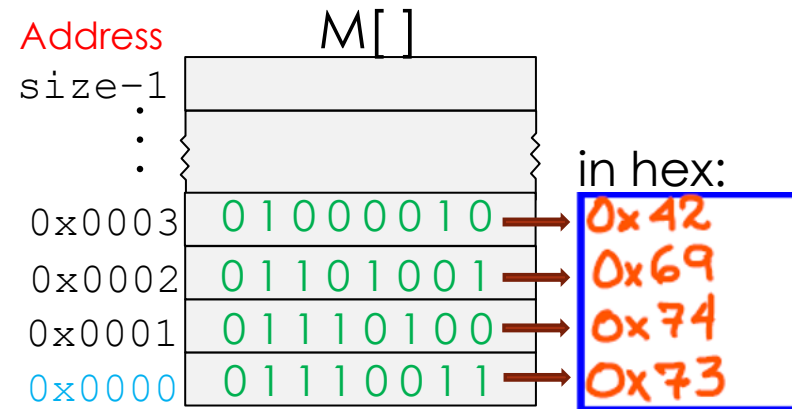
SOLUTION to Homework

Endian

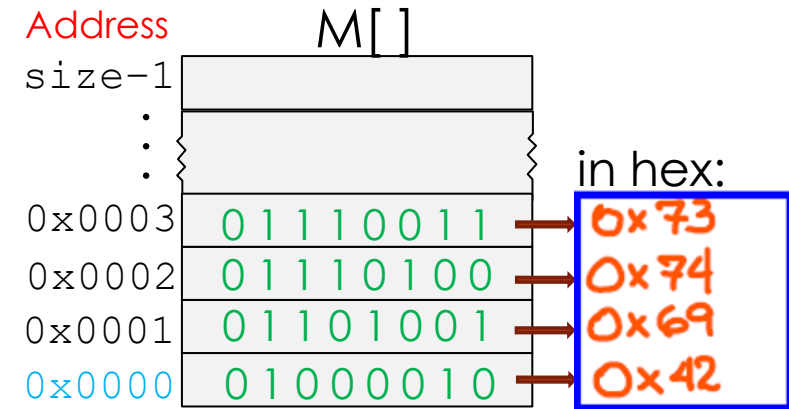
Order in which bytes stored in memory

- Convert the 4 bytes stored in memory into hexadecimal numbers and enter these hex numbers in each of the **blue** boxes.

Way 1: Little endian



Way 2: Big endian



Homework 1: Once you have done Lab 0, convert each byte stored in memory into its **hex** equivalent.

Review Let's try some Boolean algebra!

- Operations applied **bitwise** -> to each pair of bits located at same position in two operands

Homework 2 Spot the error(s):

$$\begin{array}{r} 01101001_2 \\ \& 01010101_2 \\ \hline 01000001_2 \end{array}$$

$$\begin{array}{r} 01101001_2 \\ | 01010101_2 \\ \hline 01111101_2 \end{array}$$

$$\begin{array}{r} 01101001_2 \\ \wedge 01010101_2 \\ \hline 00111110_2 \end{array}$$

$$\begin{array}{r} \sim 01010101_2 \\ \hline 10101010_2 \end{array}$$

↑
error
should be 0

Homework 3

Let's try some Shift operations!

$w=8$

1. Left Shift: ~~1011~~1001₂ << 4
10010000₂

2. Left Shift: ~~1~~0111001₂ << 1
01110010₂

3. Right Shift: 001110~~01~~₂ >> 3
logical
00000111₂

4. Right Shift: 101~~11001~~₂ >> 5
arithmetic
1111101₂

5. Right Shift: 101110~~01~~₂ >> 2
logical and arithmetic
00101110₂ 11101110₂